Feedforward Neural Networks & Backpropagation

Machine Learning â¢ Unit 2: Neural Networks & Deep Learning â¢ Study Notes

Perceptrons, multi-layer networks, activation functions, and computational graphs.

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1. Architecture & Activation Functions

Multilayer Perceptron (MLP) with hidden layers and universal approximation theorem.

Activation functions: Sigmoid, Hyperbolic Tangent (Tanh), and ReLU.

Addressing vanishing and exploding gradients with initialization schemes.

2. Backpropagation Algorithm

Forward pass: computing layer activations and intermediate cache values.

Reverse automatic differentiation using the multivariate chain rule.

Weight update optimization: Momentum, RMSProp, and Adam optimizers.